



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

SE 423: Introduction to Mechatronics

Lecture 15: Coordinate Transforms & Dead-reckoning

Marius Juston

Wednesday, March 11th, 2026

Talk to the person next to you and answer these questions.

- Sensors have noise, including the IMUs, what happens when you compute the position from the IMU?
- Why would one use a LiDAR or such a sensor for navigation?
- Stand up and with your partner and:
 - Agree on a target position and a starting point
 - Discuss ways how one would go from point A to B as accurately as possible with their eyes closed.
 - One of you closes your eyes (the partner should keep the other safe during the activity) and performs the strategy to try to get from point A to B. The other partner should not provide information during that time (Dead-Reckoning).
 - Now redo the task but now your partner can only tell you distance to the target as well (type IR sensor).

Spring Break next week, no office hours!

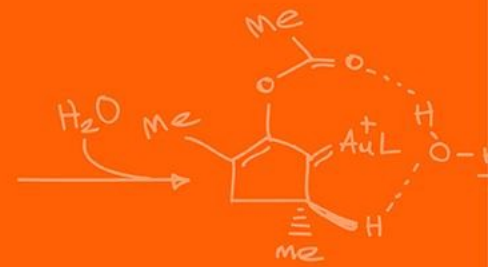
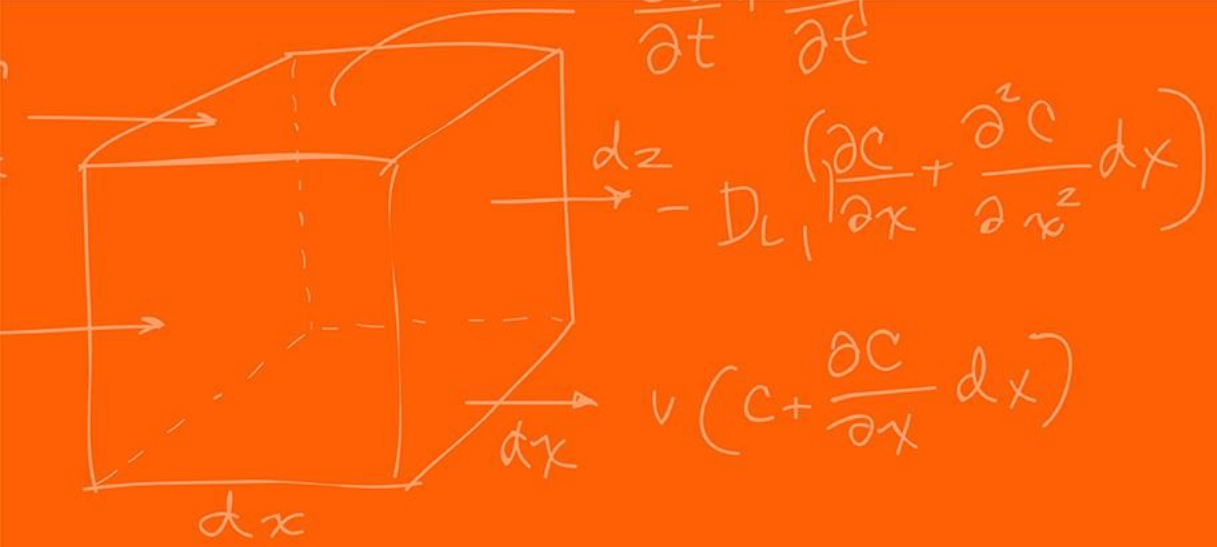
Lab: Starting Lab 5 this week. This a 1-week lab!

Homeworks:

- Check-offs for [homework 3](#) Ex 5 are due **March 24, 5PM (The end of office hours)**
- Answers and code submissions for [homework 3](#) are due **March 25, 9AM**
- Check-offs for [homework 4](#) Ex 2 and 3 are due **March 31, 5PM (The end of office hours)**

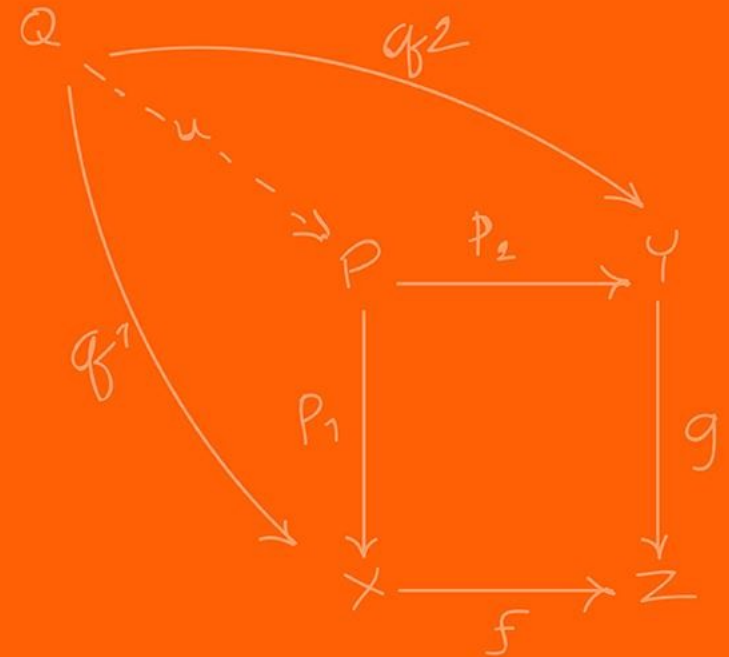
Questions?

Homework, Lab, LabVIEW?



Localization

(A brief introduction.)



We now have sensors to figure out where obstacles are. We have sensors to figure out the internal state of the robot.

We also know how to make a robot drive straight or follow a wall using feedback control.

However, if we ask the robot to “go to coordinates (5, 5) on the map,” how does it know where it currently is?

This is the fundamental question of **Localization**!

To solve localization, we need to understand that the world is not perfect and not a single sensor gives the exact solution.

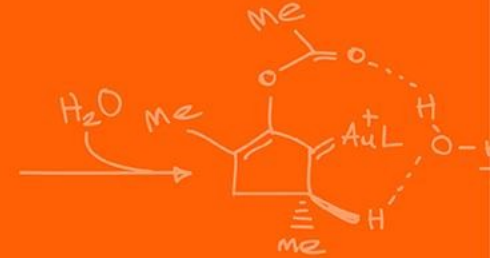
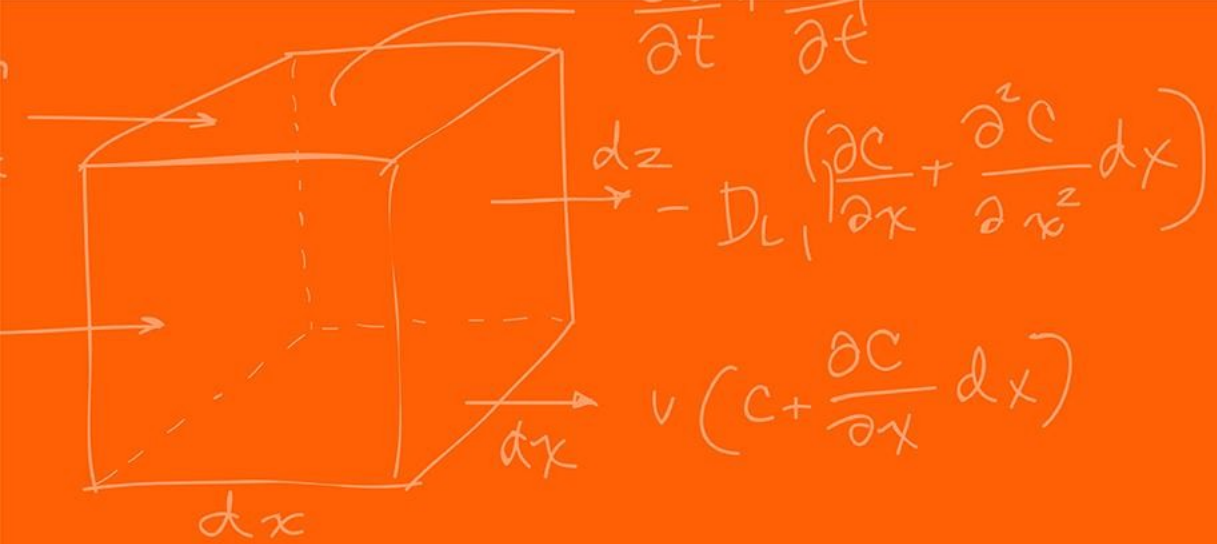
We first estimate movement using internal sensors (Dead-Reckoning).

What is a problem with just using internal sensors?

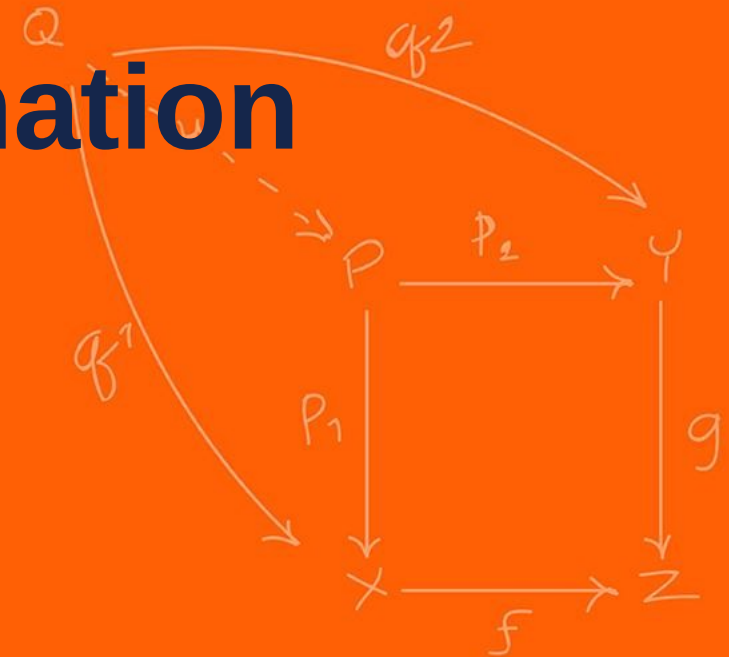
The internal sensors drift and accumulate errors!

Think of this as walking blind and relying on just counting your steps as you did at the beginning of the lecture.

Finally, we use external sensors (LiDAR, Cameras) to look at landmarks and correct these errors (SLAM)



Coordinate Transformation



A robot experiences the world from its own perspective, which we call the **Local Frame** (or Body Frame).

The map or the room exists in a fixed, global reference frame, called the **Global / World Frame** (or Inertial Frame).

When driving a car, your local reference frame is your point of view inside the car, only you have that viewpoint, and the global reference is the shared viewpoint, which is provided by the GPS map.

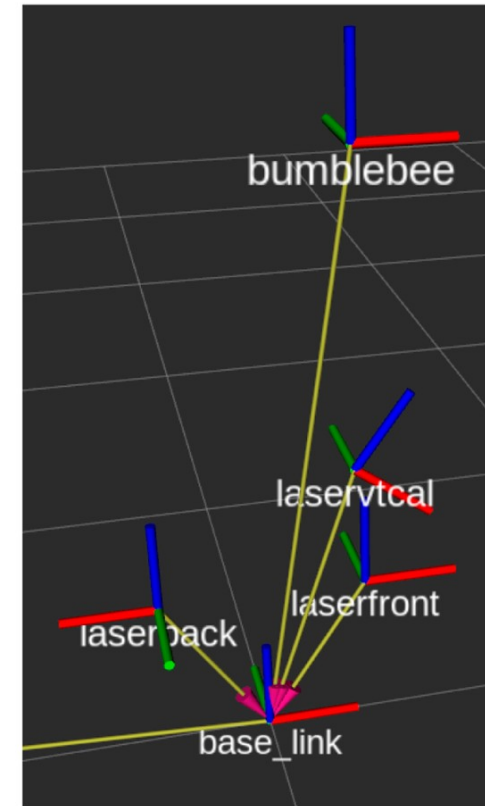
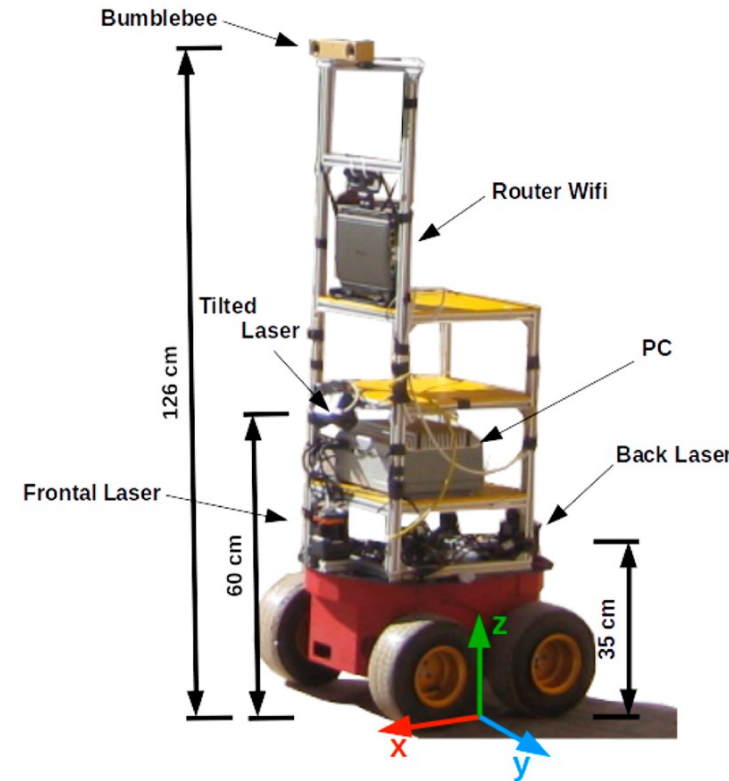
Each of the sensors view its information in its own Local Frame; however, to navigate a room, we must translate the local sensor data into the global map!

The local frames of each of the sensors can be of different positions and orientation than the robot's **body frame**.

The **body frame** is typically the frame where you define the (x, y, θ) of the robot.

So, we need to translate the sensor frames to the body frame, and then the body frame to the global frame.

If the robot to the right moves “forward 1 meter”, does its local x coordinate increase or it's y? What about it's world x, y coordinates?



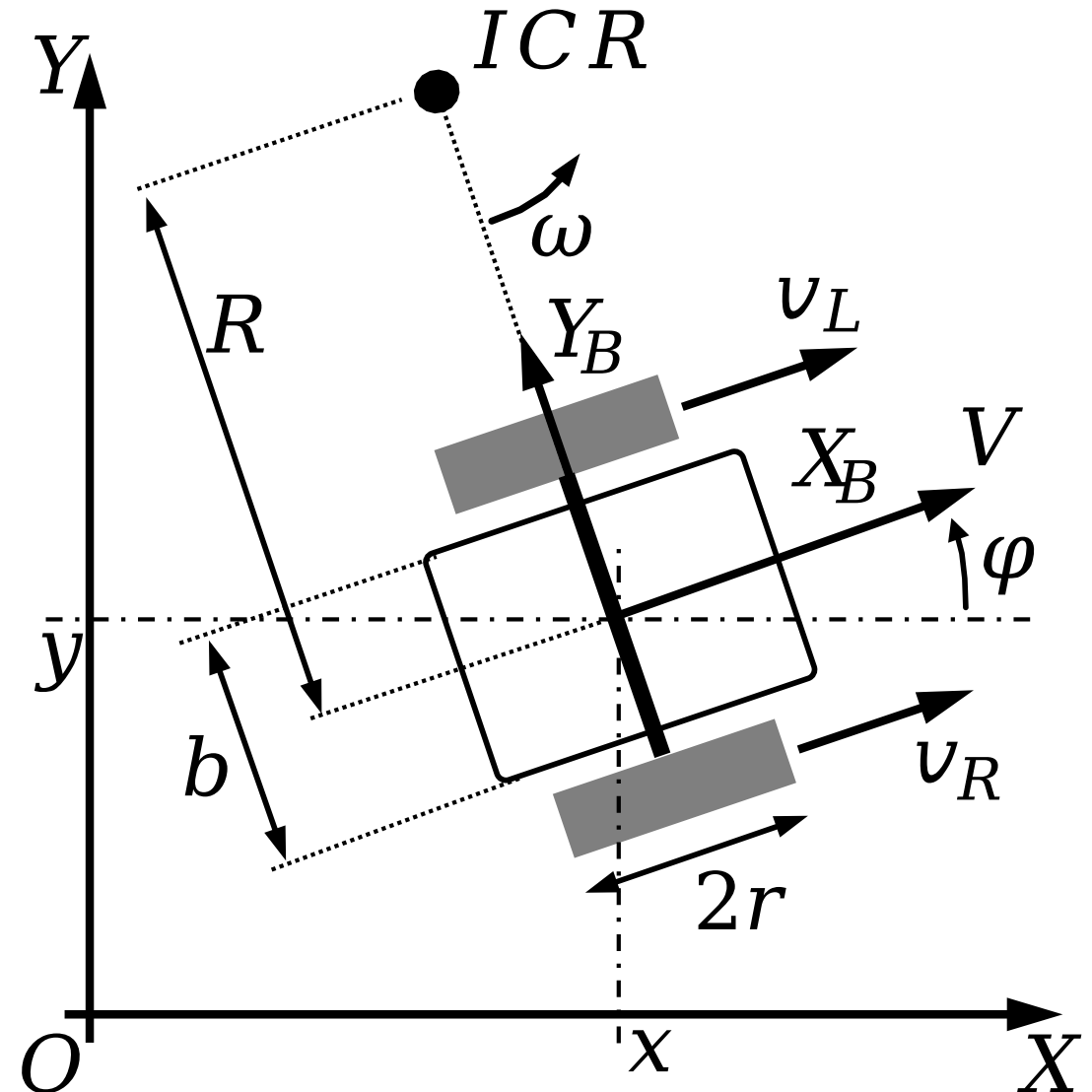
<https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2025.1671673/full>

Let the **World Frame** be defined by axes X, Y with origin O .

Let the **Body Frame** be defined by axes X_B (pointing forward) and Y_B (pointing left) with origin O_B .

The robot's pose in the world is defined by its position and its heading angle.

$$p = (x, y, \theta)$$



https://en.wikipedia.org/wiki/Differential_wheeled_robot

If the robot's heading $\theta = 0$, its axes are perfectly aligned with the world axes.

Thus, moving from the robot frame to the world frame is simply a vector addition!

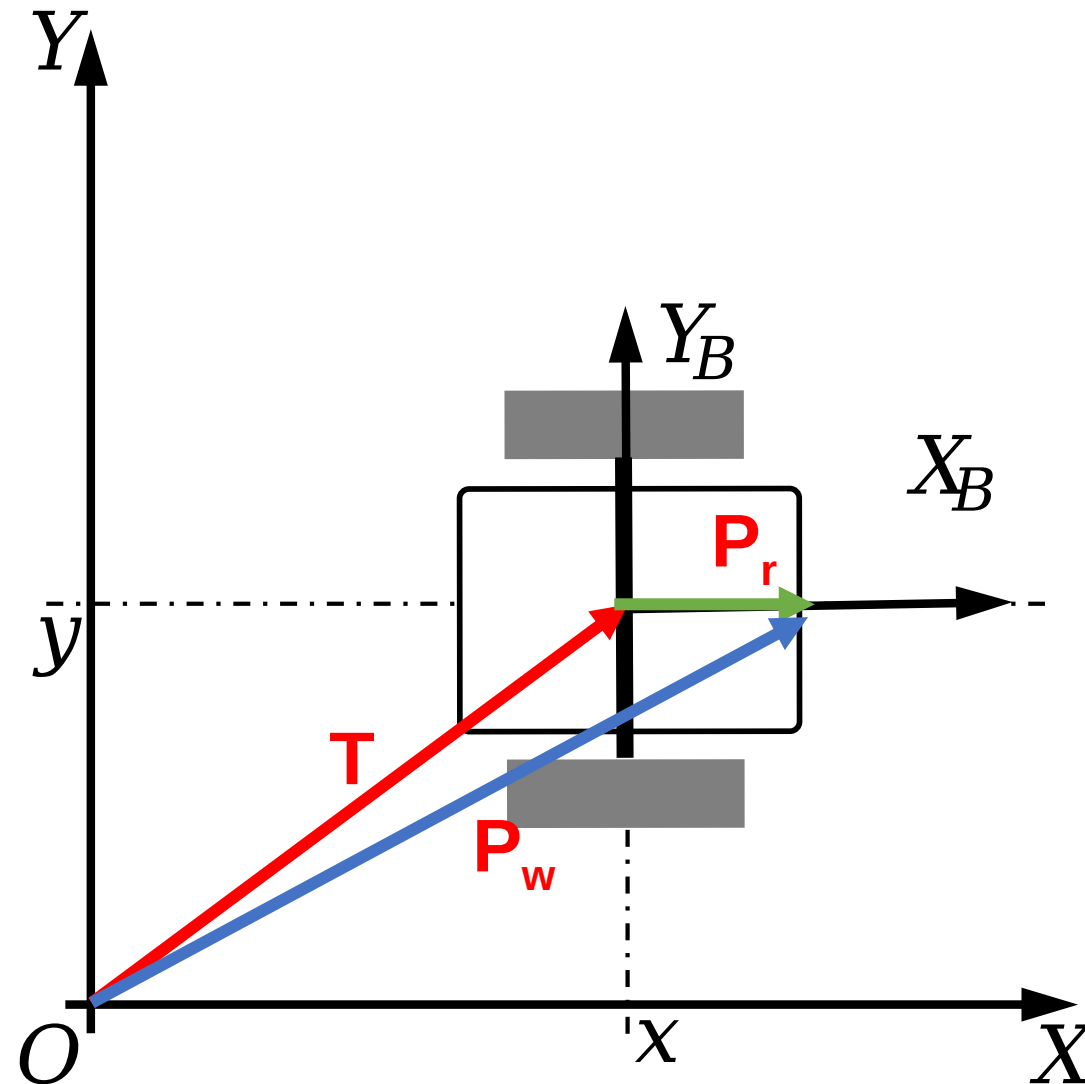
If the robot is at world coordinates $T = (x, y)$, a point $P_R = (x_r, y_r)$ in the robot frame is:

$$P_W = (x_r + x, y_r + y)$$

In the world frame. In vector math, this is represented as:

$$P_W = P_R + T$$

Translation is linear and simple; however, robot also rotate.



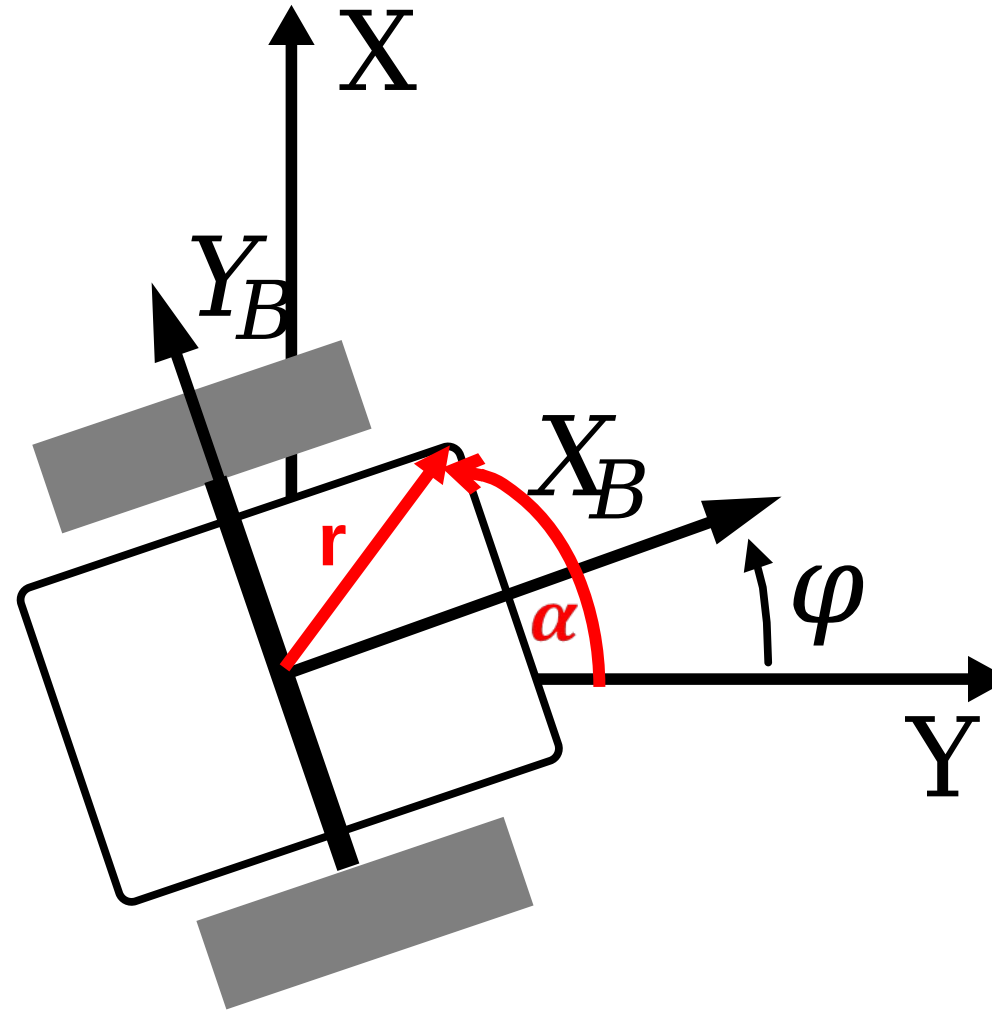
Suppose the robot frame is at the world origin but rotated by an angle φ .

We want to map point P, from the rotated frame to the fixed frame.

Let point P have polar coordinates (r, α)

Therefore,

$$(x_r, y_r) = (r \cos(\alpha), r \sin(\alpha))$$



In the world frame, the angle of point P is now at $(\alpha + \varphi)$.

The world coordinates are thus,

$$P_w = (r \cos(\alpha + \varphi), r \sin(\alpha + \varphi))$$

We can expand the following using standard trigonometric sum identities:

$$\begin{aligned}x_w &= r(\cos(\alpha) \cos(\varphi) - \sin(\alpha) \sin(\varphi)) \\y_w &= r(\sin(\alpha) \cos(\varphi) + \cos(\alpha) \sin(\varphi))\end{aligned}$$

Substituting back in the location position:

$$(x_r, y_r) = (r \cos(\alpha), r \sin(\alpha))$$

To the global position:

$$\begin{aligned}x_W &= r(\cos(\alpha) \cos(\varphi) - \sin(\alpha) \sin(\varphi)) \\y_W &= r(\sin(\alpha) \cos(\varphi) + \cos(\alpha) \sin(\varphi))\end{aligned}$$

We thus get:

$$\begin{aligned}x_W &= x_R \cos(\varphi) - y_R \sin(\varphi) \\y_W &= x_R \sin(\varphi) + y_R \cos(\varphi)\end{aligned}$$

Which is elegantly expressed using matrix multiplications!

The system of equations can be written as a 2x2 matrix $P_w = R(\varphi)P_r$

$$\begin{bmatrix} x_w \\ y_w \end{bmatrix} = \begin{bmatrix} \cos(\varphi) & -\sin(\varphi) \\ \sin(\varphi) & \cos(\varphi) \end{bmatrix} \begin{bmatrix} x_r \\ y_r \end{bmatrix}$$

The matrix $R(\varphi)$, belongs to the Special Orthogonal Group $SO(2)$.

Given that the determinant represents how much the space gets scaled after the transformation. What is the determinant of rotation matrices without doing any calculation?

It should be 1, meaning it preserves the scale and area! The robot doesn't stretch as it turns!

In reality, a robot is both translated and rotated!

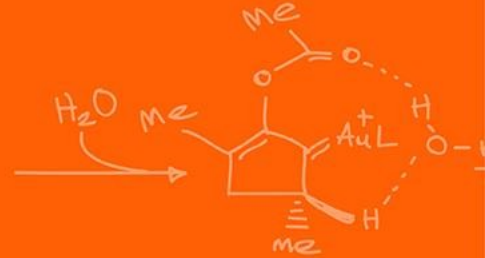
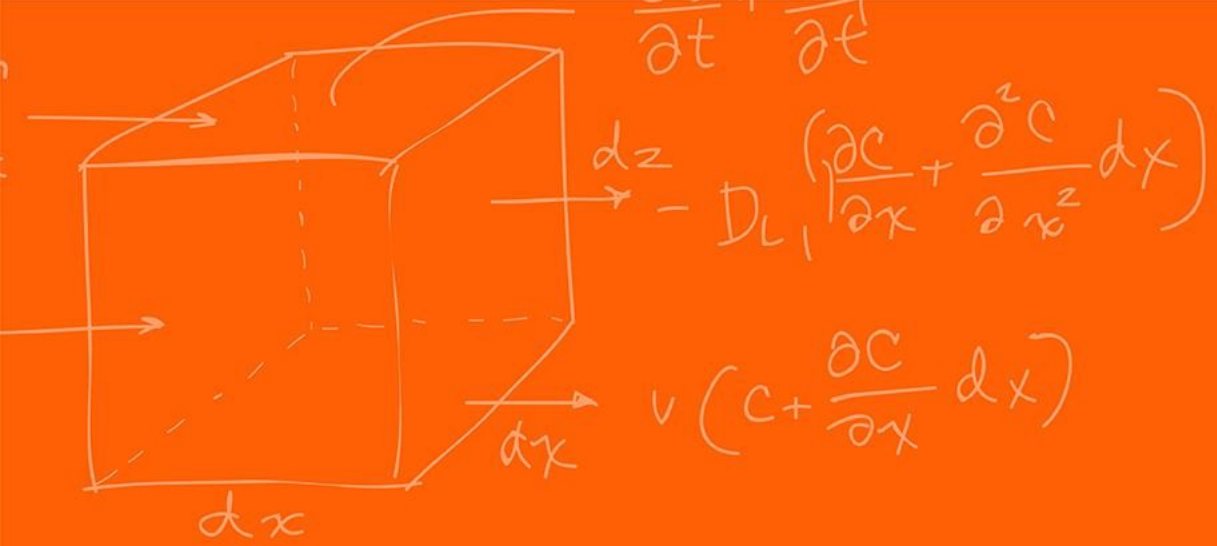
Thus, the complete transformation is:

$$P_w = R(\varphi)P_r + T$$

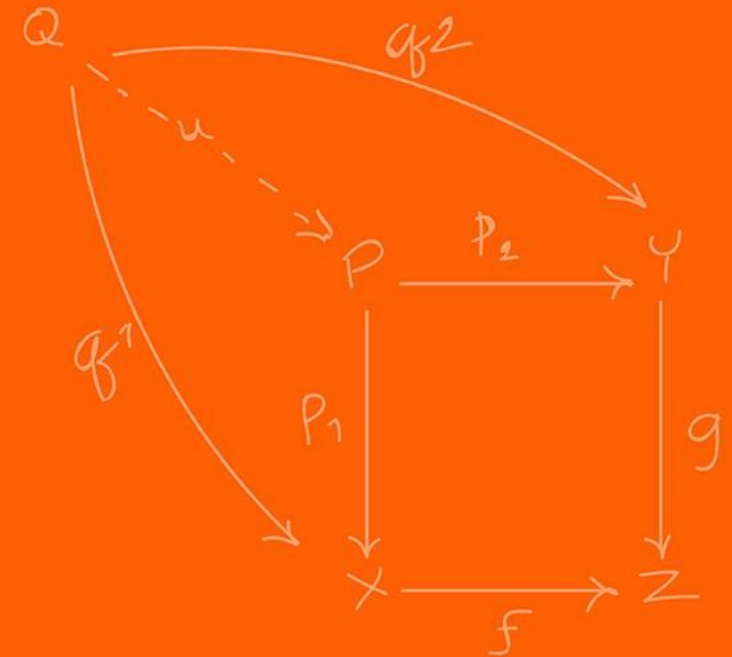
We can, however, make this into a single matrix operation, through Homogeneous Coordinates,

$$\begin{bmatrix} x_W \\ y_W \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & t_x \\ \sin(\theta) & \cos(\theta) & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_R \\ y_R \\ 1 \end{bmatrix}$$

This is the standard representation used in all modern software!



Dead-Reckoning



Now that we can map local coordinates to the world, how do we track our movement?

Dead-Reckoning (Odometry) is the process of calculating current position by integrating previously determined positions and estimated speeds over time.

It relies purely on internal sensors, primarily wheel encoders and IMUs.

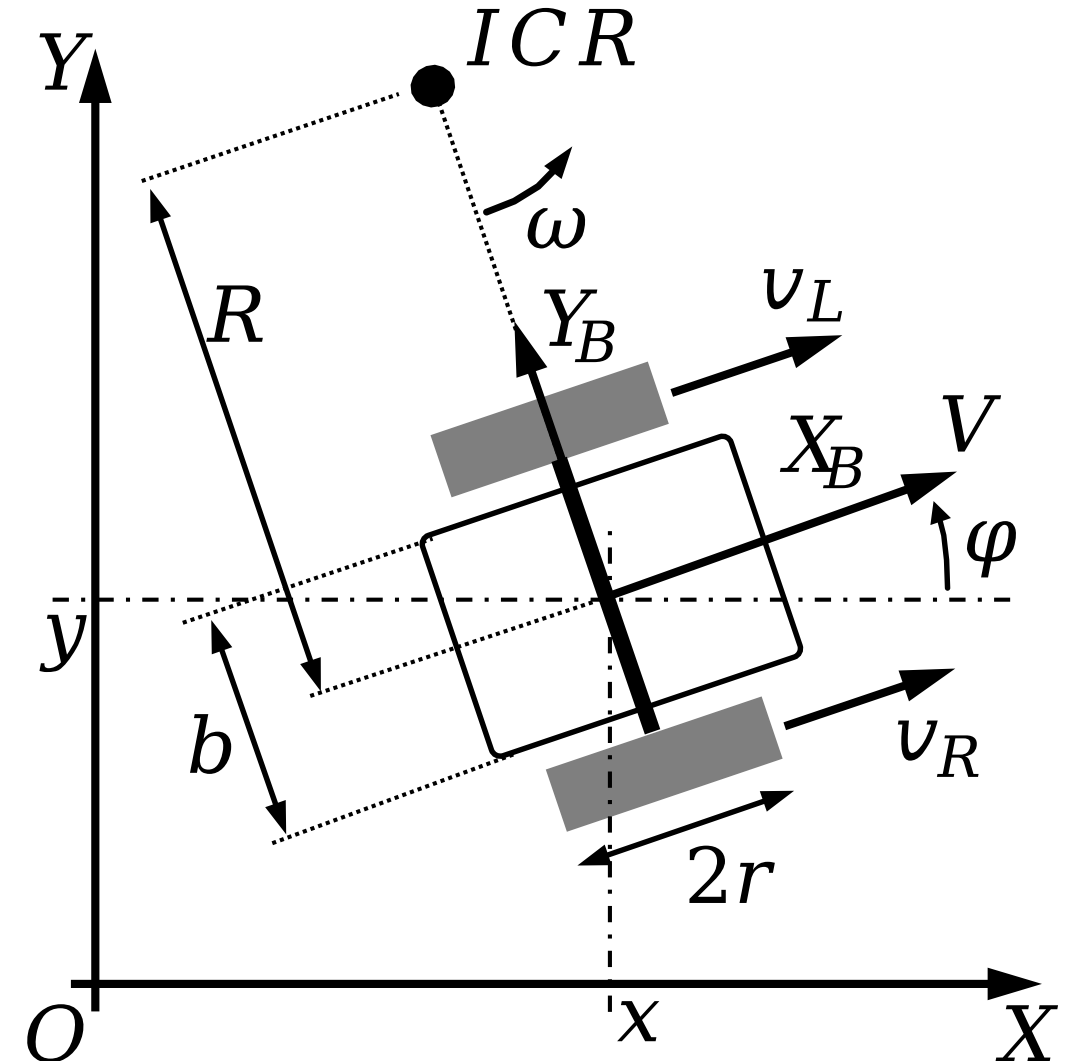
Our robot uses a differential drive system (2 independently driven wheels).

The robot's overall forward linear velocity V is the average of the two wheels:

$$V = \frac{v_L + v_R}{2}$$

If $v_L = v_R$, what is the robot doing?

The robot goes forward!

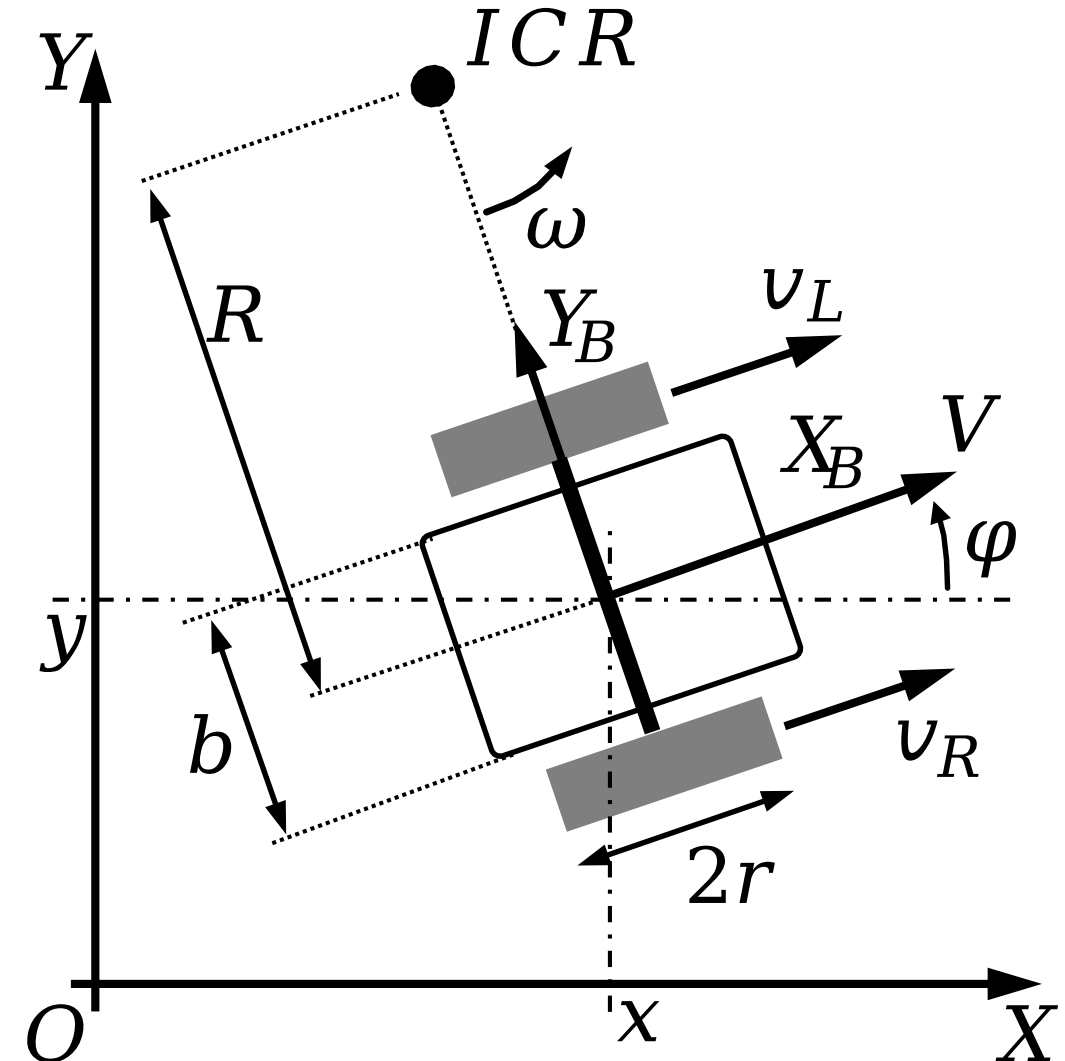


The robot's angular velocity ω depends on the difference between the wheel speeds.

$$\omega = \frac{v_L - v_R}{L}$$

If $v_L = -v_R$, what is the robot doing?

The robot rotates in place (a zero-radius turn).



Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

IMAGE / GRAPHIC

Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

IMAGE / GRAPHIC

Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

IMAGE / GRAPHIC

IMAGE / GRAPHIC

IMAGE / GRAPHIC

Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

IMAGE / GRAPHIC

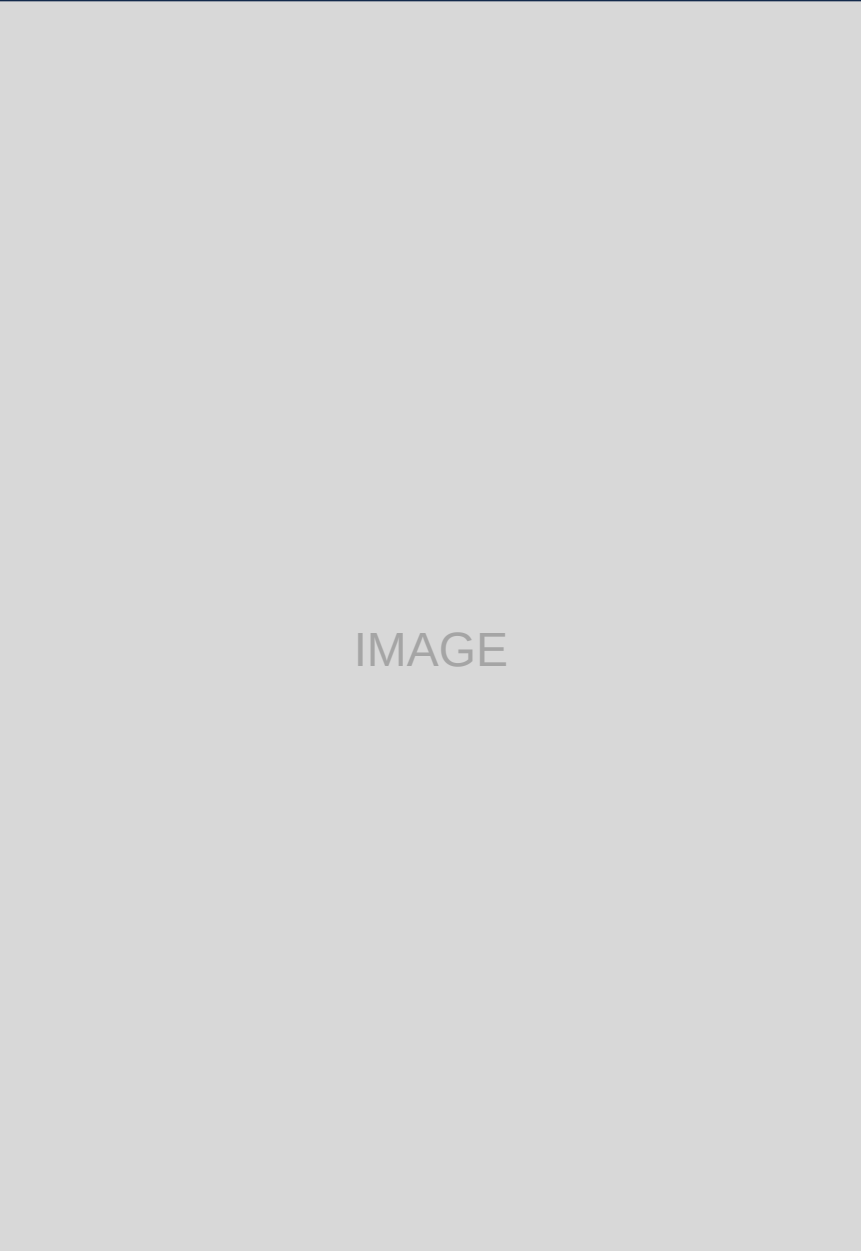


CHART / GRAPH

(Short commentary or relevant information about the contents of the chart/graph.)

IMAGE (Send to back so Block I is not covered.)

(Text here relevant to the photo for a larger call out on a given idea or message.)
(Keep this text simple and short on one or two lines.)



IMAGE

Lorem ipsum dolor sit amet consectetur adipiscing elit

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

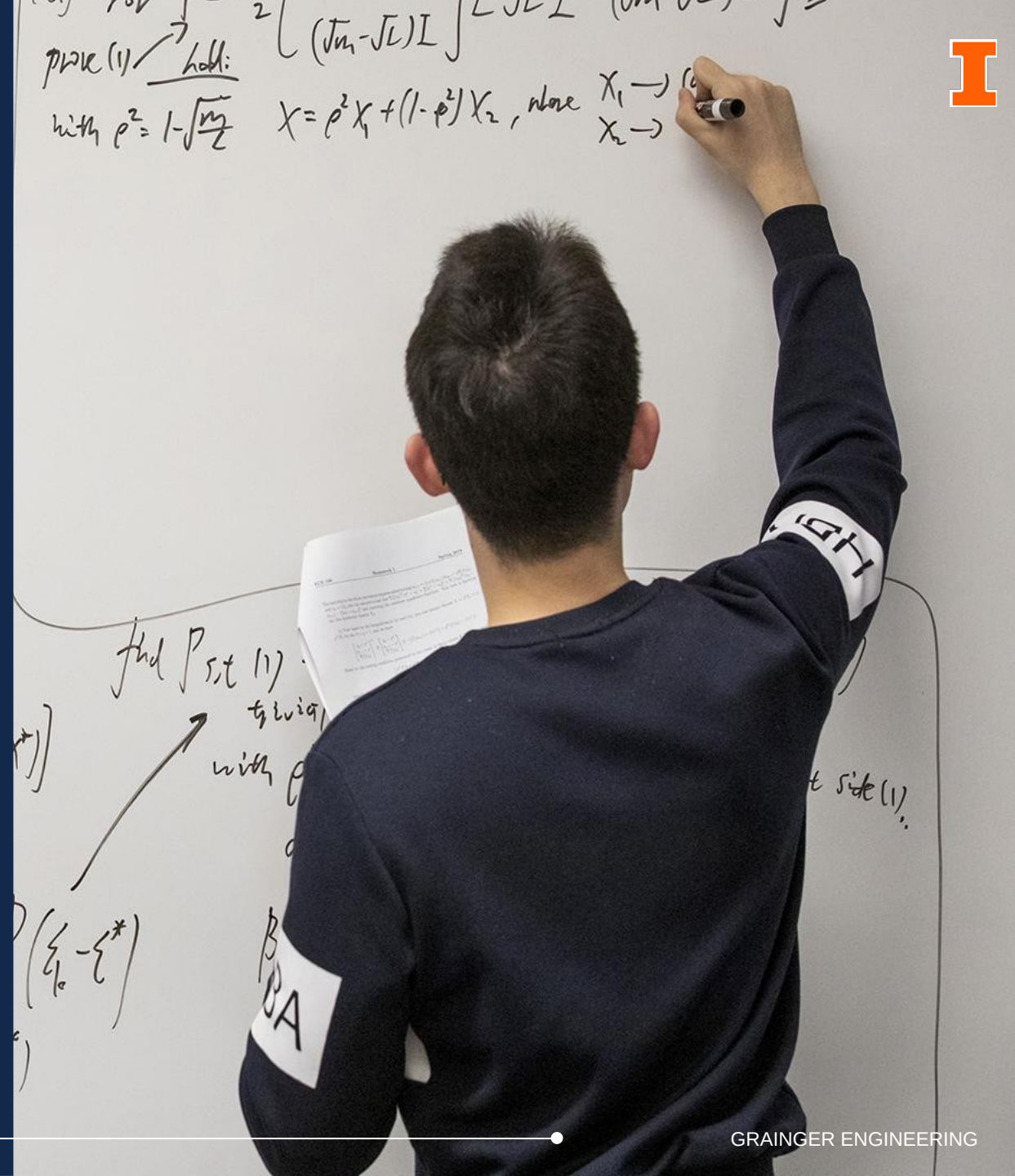
Lorem ipsum dolor sit amet consectetur adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

IMAGE

(Use a photo for the entire slide like the example shown here. Send the photo to the back so the Block I and footer text is not covered.)

(Slide the blue box and the text left or right to fit over your background image. Use this text box for a call out or caption to the image.)



(Closing slide option. Suggestions for last slide)

Summary

Thank You

Questions

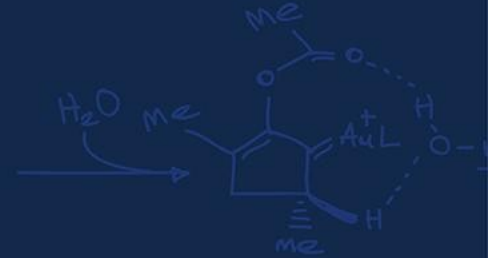
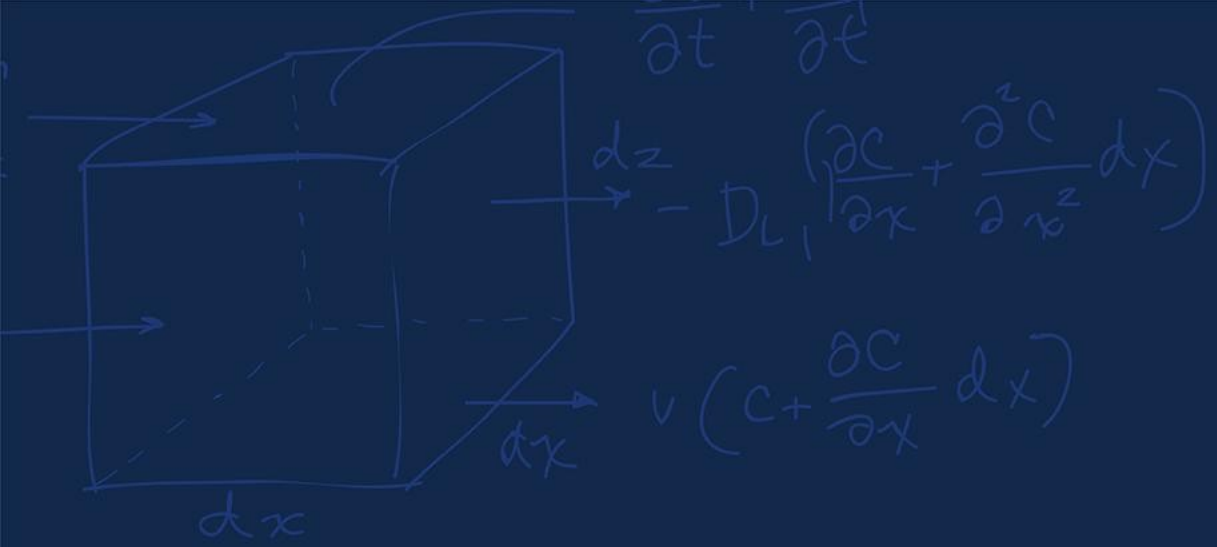
Contact Information

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

HEADING

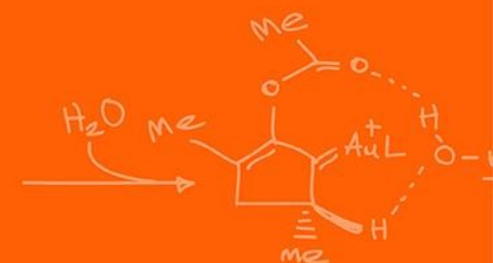
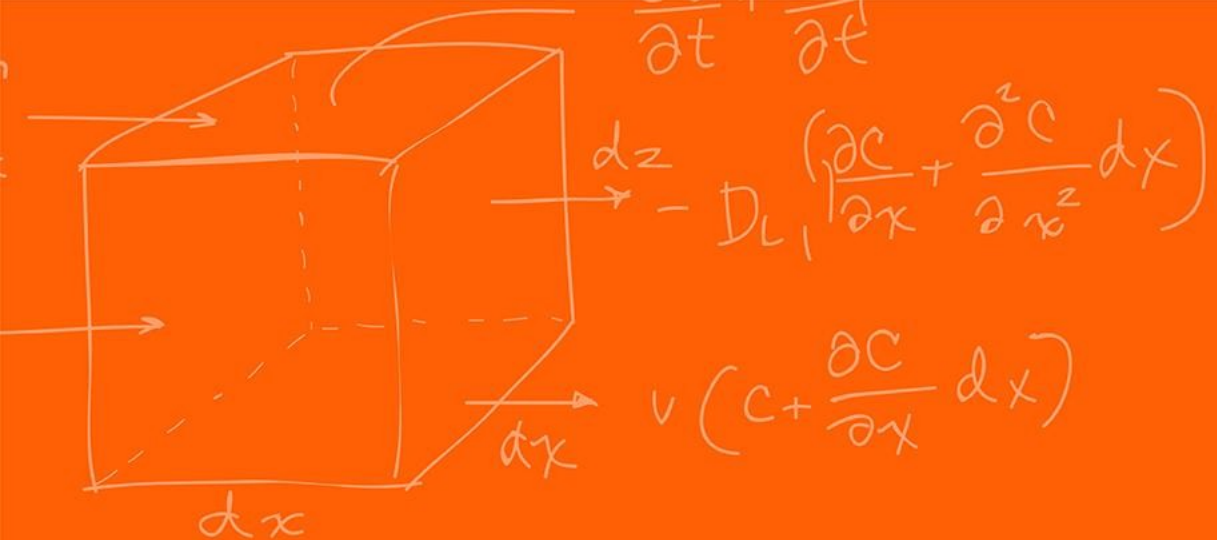
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam ut orci condimentum, commodo dui quis, faucibus mauris. Praesent nec sapien sed nunc sollicitudin.

(Main section divider slide option one.)



Questions?





The Grainger College of Engineering

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

